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Folded, Not Forgotten

Seeds, Cycles and the Persistence of Information

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Abstract. We examine a four-part thesis about the universe, which we call the *Fold Hypothesis*. (1) It began from a *seed*, an initial state that fixes everything that follows. (2) Whoever knew the seed could predict everything. (3) Free will is therefore an illusion. (4) The universe runs in cycles, each ending as black holes turn into white holes, and every trace of what happened in one cycle is folded into the seed of the next, so that each signature lasts for the entire time of the universe, like the small portion of each batch of candy said to go into the next. We state each claim precisely and test it against established physics, then derive what would have to be true for it to hold. Claim 1 is a live option: deterministic readings of quantum mechanics exist, though standard quantum mechanics is not deterministic. Claim 2 holds in principle. Given the exact seed and the exact laws, the future is a computation, and running it reproduces history the way a backtest does. What it demands is exactness and room. Because of chaos, a seed known to d digits rather than exactly predicts only about $d \cdot \ln 10/\lambda$ steps ahead. Storing the universe particle by particle would take more bits than the universe holds, unless the seed is simple. And no predictor inside the universe can predict everything, because everything includes itself. So the claim is true in principle and out of reach in practice, partly for technological reasons and partly for physical ones. Claim 3 follows for libertarian free will if Claim 1 holds, but not for the compatibilist kind. And even an exact prediction of a choice would have to compute the deliberation that is the choice, so deliberation remains the process by which outcomes get decided. For Claim 4 we prove a consistency result. Endless cycles, entropy carried across cycles, and signatures that last forever are compatible only if the cycle-to-cycle map preserves information exactly (unitarity) and the entropy that is carried over or reset is coarse-grained. If any fraction of the seed is replaced each cycle, a signature decays as p^n after n cycles: never quite zero, but soon negligible. A reversible toy universe shows the alternative: a signature scrambled beyond recognition, fully recoverable at every cycle, and returning exactly by recurrence. Unitarity is also what most physicists now expect of black holes. The Fold Hypothesis therefore survives in a sharp form, as a claim that the universe's dynamics lose no information, and we list observations that could refute its cosmological parts.

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1. Introduction

Every number in the companion paper on “every X minutes” claims (Gottfried 2026) was produced by a simulation started from a single integer, 20261003. Anyone who runs the code with that seed obtains the same tables to the last digit. In that sense the seed contains them all. And yet nobody, the author included, knew those numbers until the code had run. The seed determined them; it did not reveal them.

This paper takes that gap between determining and revealing as its starting point, and applies it to a thesis about the whole universe, which we call the *Fold Hypothesis*:

- **T1 (seed).** The universe began from an initial state, the seed, that fixes everything that happens afterwards.
- **T2 (prediction).** Whoever knew the seed could predict the trajectory of every particle for the rest of time.
- **T3 (free will).** Free will is therefore an illusion.
- **T4 (cycles and persistence).** The universe repeats in cycles, of order 110 billion years each, ending as black holes collapse into white holes. What happened in a cycle is not forgotten but folded into the next seed, so every signature lasts for the entire time of the universe. A widely repeated account holds that each batch of Tootsie Rolls includes a little of the previous batch; T4 says the universe is made the same way. Nothing is thrown away: it is folded in.

Versions of each claim have long histories: Laplace’s demon for T1–T2 (Laplace 1814), hard determinism for T3 (van Inwagen 1983), and oscillating and cyclic cosmologies for T4 (Tolman 1931; Tolman 1934; Steinhardt and Turok 2002; Baum and Frampton 2007; Penrose 2010; Haggard and Rovelli 2015; Popławski 2010; Smolin 1997). Our aim is not to endorse or dismiss the thesis but to make it precise enough to be wrong. Section 2 separates determinism from predictability. Section 3 asks what follows for free will. Section 4 reviews how a universe could cycle and what limits entropy places on cycling. Section 5 introduces a model of carry-over between cycles, proves when a signature can last forever, and illustrates the answer with a toy universe. Section 6 lists observations that could refute the cosmological parts. Section 7 gives a verdict on each claim.

2. Seeds: determinism without prediction

2.1 The seed picture

Write the state of the universe at time t as $x(t)$ and its laws as a flow Φ_t , so that $x(t) = \Phi_t(s)$, where $s = x(0)$ is the seed. T1 says that such a Φ exists and is single-valued. T2 says, further, that someone could evaluate $\Phi_t(s)$ before time t arrives. Laplace stated both in one sentence (Laplace 1814): an intellect that knew all forces and all positions “would embrace in the same formula the movements of the greatest bodies of the universe and those of the lightest atom; for it, nothing would be uncertain.” A pseudorandom number generator is a working model. Its entire output is a fixed function of the seed, yet the output looks random to anyone who does not hold the seed, and computing it means running the generator.

2.2 Is physics deterministic?

Classical mechanics is deterministic, and so is general relativity for globally hyperbolic spacetimes. Quantum mechanics, in its standard formulation, is not: measurement outcomes are random with probabilities given by the Born rule. Bell’s theorem shows that no local theory of predetermined outcomes reproduces quantum correlations (Bell 1964). Still, three readings of quantum mechanics are fully deterministic, and each gives T1 a different form. In Everett’s interpretation, the universal wavefunction evolves unitarily and deterministically from its initial state, which is the seed (Everett 1957). Every outcome occurs on some branch, but an observer cannot predict which branch they will find themselves on. In Bohmian mechanics, particle positions follow deterministic trajectories, but their initial values can be known only up to the $|\psi|^2$ distribution, which makes the uncertainty absolute rather than practical (Dürr et al. 1992). In superdeterministic models, including ’t Hooft’s cellular-automaton interpretation (’t Hooft 2016), even the choice of what to measure is fixed by the initial state, so Bell’s theorem does not apply. Superdeterminism is the version closest to T1 as stated, and also the most contested. Our verdict is that T1 is a coherent and live option, but it is not established. It requires one of these interpretations, and current experiments do not distinguish among them.

2.3 What it takes to predict from the seed

Grant T1. Then T2 is true in principle, and the opening example shows how. Given the seed 20261003 and the code, anyone can reproduce every number in the companion paper, before or after it was published. Given the universe’s exact seed and exact laws, the same procedure would reproduce its whole history. Run over the past, it is a backtest that recovers what happened; run past the present, it is a forecast. Nothing in physics forbids the computation itself. What T2 needs is three things, each harder than it looks.

An exact seed. The seed must be known exactly, not approximately.

Proposition 1 (precision horizon). *In a system whose largest Lyapunov exponent is $\lambda > 0$, keeping a forecast within tolerance δ for time t requires knowing the seed to within $\varepsilon \lesssim \delta e^{-\lambda t}$, that is, about $\lambda t / \ln 2$ bits per unstable direction beyond the tolerance. A seed known to d decimal digits supports forecasts of about $d \ln 10 / \lambda$ time units.*

Proposition 1 is not an objection to T2. It says how exact the seed must be. With the seed known exactly, as in the companion paper’s simulations, chaos costs nothing, and the computation reproduces the trajectory perfectly. With any rounding at all, the forecast fails after a time that grows only with the logarithm of the precision. Table 1 checks Proposition 1 on Arnold’s cat map (Arnold and Avez 1968), the simplest chaotic system, for which $\lambda = \ln[(3 + \sqrt{5})/2] = 0.962$. Each additional digit of the seed buys about 2.4 more steps. Fifteen digits, the precision of ordinary floating-point arithmetic, buy 34. Real matter is far more sensitive. By Berry’s estimate, the gravitational pull of a single electron at the edge of the observable universe is enough to make the motion of gas molecules unpredictable after a few dozen collisions (Berry 1978). Chaos in the weather limits forecasts in the same way (Lorenz 1963).

Seed precision (decimal digits)	3	6	9	12	15
Steps until the forecast fails (simulated)	6	13	20	27	34
$d \ln 10/\lambda$ (Proposition 1)	4.8	12.0	19.1	26.3	33.5

Table 1. The precision horizon of the cat map. A forecast fails when its error exceeds 0.1 on the unit torus. Two seeds differing by 10^{-d} diverge after about $2.4d$ steps.

Quantum mechanics raises a further question: whether an exact seed can even be known. Under Bohmian mechanics the initial positions are definite but cannot be measured beyond the $|\psi|^2$ distribution (Dürr et al. 1992). Under Everett the exact seed is the wavefunction, and running it forward predicts every branch at once, but not which one an observer will find themselves on (Everett 1957).

Room to store it and time to run it. The observable universe can have performed at most about 10^{120} elementary operations on at most about 10^{90} bits since the Big Bang (Lloyd 2002). A molecule in a gas collides some 10^{10} times a second, and each collision consumes a few bits of initial precision. Writing down each of the universe’s roughly 10^{80} particles precisely enough to predict a single year would therefore take something like 10^{97} bits. This is a limit of physics, not of current technology: no computer built from part of the universe can store more than the universe holds. There is one escape, and it favours T1. The seed need not be large. If the universe began in a simple state, as a pseudorandom generator begins from a single integer, its exact description could be short, and what would grow is only the work of running it forward. Whether that work could be done faster than the universe does it is open. For chaotic dynamics no general shortcut is known, and Wolfram has conjectured that most such processes are computationally irreducible (Wolfram 2002).

A predictor outside what it predicts. A predictor that is part of the universe must also predict itself, including the effect of announcing its own prediction. Wolpert proved that no physical inference device can correctly predict every other such device, or every fact about the universe that contains it (Wolpert 2008). The argument is diagonal, like the halting problem. It does not apply to a predictor outside the universe, or to an inside predictor forecasting systems it does not influence. It applies to “everything,” because everything includes the predictor.

Who predicts	Can the seed predict everything?
An observer outside the universe, with the exact seed, the exact laws and an unlimited computer	Yes, if T1 holds: the past as a backtest and the future as a forecast
A predictor inside the universe, with the exact seed and laws	Everything except its own future and whatever responds to its predictions; and it can run ahead of real time only if the seed is compact and a computational shortcut exists
Current technology	No: the seed is unmeasured, the laws are incomplete (there is no accepted theory of quantum gravity), and the computation is far beyond any existing machine

Table 2. Prediction from the seed at three levels.

So T2 is true in principle and out of reach in practice. Part of the gap is technological: we do not know the seed or the final laws, and our computers are small. Part of it is physical: exactness, capacity and self-reference. No simulation running inside the universe can contain itself, which is why the universe may be its own fastest simulation, and is certainly the only complete one from inside.

3. Free will

If T1 holds, every choice was fixed by the seed. Van Inwagen’s consequence argument puts the libertarian worry exactly: if our acts are consequences of the laws of nature and events in the remote past, and neither is up to us, then our acts are not up to us either (van Inwagen 1983). In that sense T3 follows from T1. Libertarian free will, the ability to have done otherwise with the past and the laws held fixed, is an illusion in a deterministic universe.

Two qualifications keep T3 from proving more than it does. First, indeterminism does not rescue libertarian freedom: a choice settled by a quantum coin toss is no more “up to me” than one settled by the seed. The free will theorem of Conway and Kochen shows that if experimenters’ choices are not determined by the past, particle responses are not either (Conway and Kochen 2006). It ties the two kinds of freedom together without showing that either exists. Second, compatibilists define free will as acting on one’s own reasons, with the right relation between one’s desires and one’s actions, and on that definition free will is untouched by determinism (Frankfurt 1971; Dennett 1984). Neuroscience has not settled the matter. The readiness potential once taken to show that the brain decides before we are aware (Libet et al. 1983) is now widely read as accumulated neural noise rather than a hidden decision (Schurger et al. 2012).

Section 2 adds something to this old debate. Suppose an outside predictor held the exact seed and ran the universe forward. Its forecast of your choice would be obtained by computing, step by step, the same process that is your deliberation; the forecast would contain a copy of you deciding. Deliberation is therefore not idle, even in a fully predictable universe. It is the process by which the seed’s consequences get worked out, wherever they are worked out. In the language of the opening example, the seed fixes the output, but the output does not exist until the generator runs, and a person deliberating is the generator running. The defensible form of T3 is this: *the choice is determined, and you are the computation that determines it*. What is illusory is the sense that the past could have been followed by a different choice. What is not illusory is that the choice is made by you: any prediction of it has to run you first.

4. Cycles

4.1 How a universe could cycle

Cyclic cosmologies come in two families. In *global* models the whole universe ends and restarts. Tolman’s oscillating universe recollapses and re-expands (Tolman 1931; Tolman 1934); the ekpyrotic cyclic model of Steinhardt and Turok uses collisions between branes (Steinhardt and Turok 2002); Baum and Frampton’s model discards most of space at each turnaround (Baum and Frampton 2007); and Penrose’s conformal cyclic cosmology joins the infinitely expanded end of one aeon to the Big Bang of the next (Penrose 2010). In *local* models the cycle runs through black holes. Black holes may tunnel into white holes (Haggard and Rovelli 2015); a black hole’s interior may bounce into a new expanding universe, as in Einstein–Cartan cosmology with torsion (Popławski 2010); and in Smolin’s cosmological natural selection, each black hole spawns a universe whose constants are those of its parent, slightly changed (Smolin 1997).

T4’s mechanism, black holes collapsing into white holes, belongs to the local family, and that has a consequence the thesis should take on board. A universe with many black holes produces many successors, so history is a branching tree rather than a single line of cycles. Smolin’s picture is the closest existing theory to the tootsie-roll intuition: each offspring universe carries a small, slightly

mutated inheritance from its parent. But there are many offspring, and “the next seed” is one of a great many.

4.2 Will this universe end?

The universe is 13.8 billion years old (Planck Collaboration 2020). If dark energy is a cosmological constant, it will expand forever and never recollapse, and global cycles are ruled out unless, as in Penrose’s scheme, the end is redefined. Recent baryon-acoustic-oscillation measurements from DESI prefer dark energy that weakens with time, at a significance of $2.8\text{--}4.2\sigma$ depending on the data combined (DESI Collaboration 2025). If that holds, a recollapse becomes possible, and one fit estimates a total lifespan of about 33 billion years (Luu et al. 2025). T4’s figure of 110 billion years has no derivation behind it. We therefore treat the cycle length T_c as a free parameter. On T4’s value, we are about an eighth of the way through the present cycle.

4.3 Tolman’s entropy problem

Entropy grows within each cycle. Tolman noticed that if it also carries over, each cycle starts with more entropy than the last, so cycles grow longer and the sequence cannot be truly periodic (Tolman 1934). A simple recursion makes the constraint explicit.

Proposition 2 (entropy across cycles). *Let S_n be the entropy at the start of cycle n , suppose each cycle produces entropy $a > 0$, and let a fraction p of the entropy carry over: $S_{n+1} = pS_n + a$. If $p < 1$, $S_n \rightarrow a/(1 - p)$ and the cycles approach a steady state. If $p = 1$, $S_n = S_0 + na$ grows without bound.*

Every cyclic model that repeats forever therefore discards entropy somewhere. Baum and Frampton discard almost all of space (Baum and Frampton 2007); Penrose discards mass and with it the scale against which entropy is counted (Penrose 2010). T4 says that entropy carries over *and* that cycles repeat forever. By Proposition 2 the two can coexist only if the carry-over is partial, and Section 5 shows that partial carry-over means signatures fade. The way out, developed next, is that the entropy that must be shed and the information that must be kept are not the same thing.

5. Folding versus forgetting: how long does a signature last?

5.1 The model

Let s_n be the seed of cycle n , and let the seed of the next cycle be produced by a map that may lose information: $s_{n+1} = F(s_n, \xi_n)$, where ξ_n is fresh material not determined by s_n . A *signature* is anything written into s_0 , and it survives to cycle n to the extent that s_n still carries information about s_0 . We measure that with the mutual information $I(s_0; s_n)$. The *tootsie-roll map* keeps each part of the seed with probability p , passing it through a reversible scramble, and replaces it with fresh material otherwise.

Proposition 3 (persistence of a signature). *(a) If F is invertible and uses no fresh material, as unitary evolution does, then $I(s_0; s_n) = H(s_0)$ for every n : the signature lasts for the entire time of the universe. (b) Under the tootsie-roll map with carry-over p , applied independently to each part, $I(s_0; s_n) = p^n H(s_0)$ when the fresh material is independent of the signature. (c) More generally, if each cycle is a noisy channel with strong data-processing constant $\eta < 1$, then $I(s_0; s_n) \leq \eta^n H(s_0)$.*

Part (b) captures the tootsie-roll intuition exactly. A signature never vanishes, since $p^n > 0$ for every n , but it halves every $\ln 2 / \ln(1/p)$ cycles. With 10% carry-over, a signature retains 10^{-10} of its information after ten cycles. “Never forgotten” in the literal sense of “not exactly zero” is true of any leaky universe. “Lasts the entire time,” in the sense of remaining recoverable, is true only of case (a).

5.2 A toy universe

Figure 1 runs both cases on a 101×101 grid. The signature “US” is written into the seed. Each cycle applies Arnold’s cat map, $(x, y) \mapsto (2x + y, x + y) \bmod 101$, which is a bijection of the grid and so loses nothing (Arnold and Avez 1968). In the leaky version, each cell is also replaced with probability $1 - p$ by fresh material of the same average density.

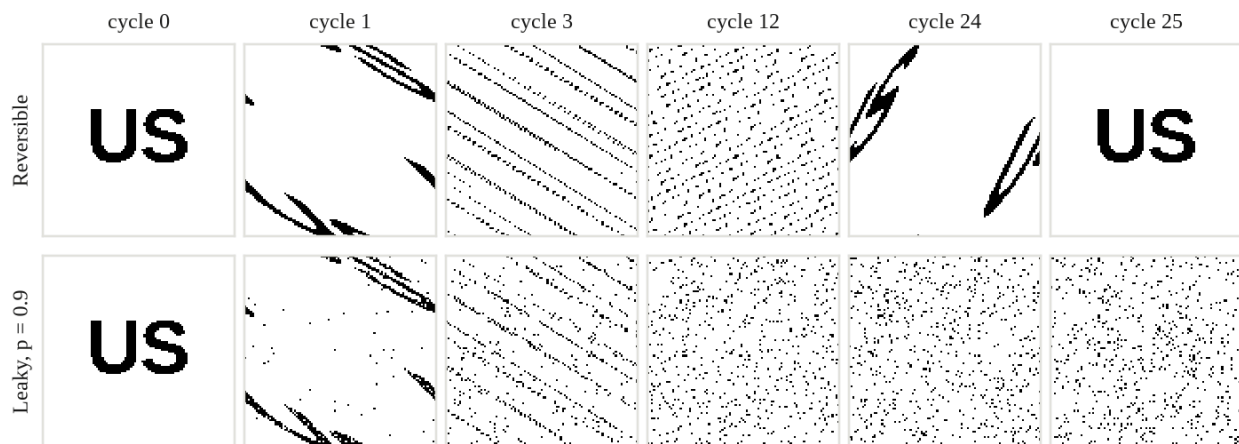


Figure 1. A signature in a reversible and a leaky universe. Top: the cat map scrambles “US” until it is unrecognizable (cycle 12), then the exact seed recurs at cycle 25. Bottom: with 90% carry-over per cycle, the scramble is the same, but fresh material accumulates and the signature never returns.

Two different questions can be asked of the signature, and Figure 2 separates them. Does cycle n *resemble* the seed? In the reversible universe, almost never: the correlation between the scrambled grid and the original stays within ± 0.07 from the second cycle to the twenty-fourth. Is the signature *recoverable* from cycle n ? In the reversible universe, always: undoing the map returns the seed exactly, at every cycle. In the leaky universe, the recoverable signature follows p^n , as Proposition 3(b) predicts (0.91, 0.82, 0.74 for the first three cycles at $p = 0.9$, against 0.90, 0.81, 0.73). The intuition that “we just get folded into the new seed” is precisely the first case: folded so thoroughly that nothing looks the same, yet nothing is lost.

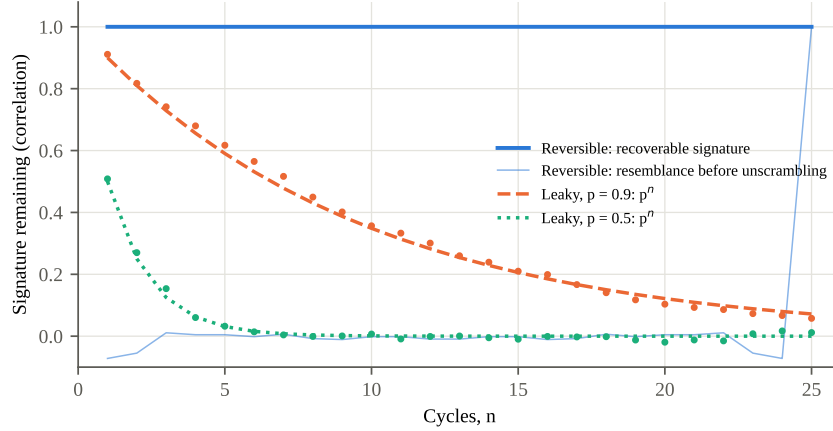


Figure 2. How long the signature lasts. Reversible universe: recoverable signature (thick line) and resemblance to the seed before unscrambling (thin line, which jumps to 1 at the recurrence). Leaky universes: recovered signature (dots) against p^n (lines) for $p = 0.9$ and 0.5 .

The toy universe also returns exactly to its seed, after 25 cycles. This is Poincaré recurrence (Poincaré 1890), which holds for any reversible system with finitely many states. For the real universe the recurrence time is of order e^S , and with an entropy of about $10^{104} k_B$ for the observable universe (Egan and Lineweaver 2010), that is incomparably longer than any 110-billion-year cycle. So cycles of T4’s length are not repeats. Each is new, though each contains, scrambled, everything that came before.

5.3 When the thesis is consistent

Theorem 1 (consistency of T4). *The three parts of T4 (cycles that repeat forever, entropy that carries over, and signatures that last for the entire time of the universe) are jointly consistent if and only if the cycle-to-cycle map is information-preserving on the fine-grained state, while the entropy that grows and is carried over or shed is coarse-grained.*

The proof is short. Signatures that last forever require case (a) of Proposition 3, since any loss rate $\eta < 1$ sends recoverable information to zero. Endless cycles with entropy carried over require $p < 1$ in Proposition 2, because otherwise entropy diverges. Both can hold only if the two p ’s measure different things. They can. Under unitary evolution, the fine-grained (von Neumann) entropy of a closed system never changes. What grows is coarse-grained entropy, our ignorance of where the information went, which is the second law as it is actually observed. A bounce can then reset the coarse-grained entropy, as Penrose’s rescaling or Baum and Frampton’s discarding would, while the fine-grained state, and every signature in it, passes through intact but scrambled.

Theorem 1 converts the cosmological part of the thesis into a claim about fundamental physics: *the universe’s dynamics, including black holes and bounces, are unitary.* That is a live question. Hawking argued that black-hole evaporation destroys information (Hawking 1975). Page showed what the radiation of an information-preserving black hole must look like (Page 1993), and Hayden and Preskill showed that such a black hole scrambles what falls in and later releases it (Hayden and Preskill 2007). Recent semiclassical calculations reproduce Page’s curve (Penington 2020; Almheiri et al. 2019), and most researchers now expect black holes to preserve information. Scrambling of this kind has been verified in small quantum processors (Landsman et al. 2019). The thesis’s claim that

“we are never forgotten, we just get folded into the new seed” is, in this light, close to the current consensus, provided that the bounce, too, is unitary.

One more connection ties this paper to its companion (Gottfried 2026). Information is not the only thing a cycling universe might remember; it might also keep time. If the length of each cycle varies independently, the phase of the cosmic cycle is forgotten geometrically even when no information is lost, just as a renewal process forgets its phase. Only a bounce whose timing errors are measured against a fixed schedule could keep cosmic time forever.

6. How the Fold Hypothesis could be refuted

A thesis worth holding should be exposed to evidence. The cosmological parts of T4 are.

- **Dark energy.** A global recollapse needs dark energy that weakens enough to stop expansion. If DESI, Euclid and the Rubin Observatory converge on a cosmological constant, global cycles of any length fail, leaving only local, black-hole versions of T4. If they confirm weakening dark energy, the implied time to recollapse is itself a measured, not assumed, cycle length to compare with 110 billion years (DESI Collaboration 2025; Luu et al. 2025).
- **Traces of a previous cycle.** Conformal cyclic cosmology predicts “Hawking points,” warm spots in the cosmic microwave background left by black holes of the previous aeon. A claimed detection (An et al. 2020) has been disputed by an independent reanalysis (Jow and Scott 2020). Future maps can settle the question.
- **Black-hole reproduction.** Smolin’s version predicts that our universe’s constants nearly maximize black-hole production. He used this to predict that no neutron star exceeds about 1.6 solar masses (Smolin 1997). The pulsar PSR J0740+6620 weighs about 2.08 solar masses (Fonseca et al. 2021), which counts against that version of the prediction.
- **Unitarity.** If the black-hole information problem were resolved in favour of genuine information loss, Theorem 1 would refute the “lasts the entire time” part of T4 outright.
- **Prediction in principle.** T2 cannot be tested by running the universe, but its ingredients can. One is a final set of laws, including quantum gravity. The other is a simple initial state, compact enough to write down; Penrose’s proposal that the Big Bang had nearly zero gravitational entropy points that way (Penrose 2010). If the early universe turns out to require an irreducibly complex seed, T2 loses its one escape from the capacity limit.

7. Verdict

Claim	Status	What would make it true, or false
T1. A seed fixes everything	Open	True under Everett, Bohmian or superdeterministic readings; false under standard quantum mechanics
T2. Knowing the seed predicts everything	True in principle; out of reach in practice	Needs the exact seed and laws, like a backtest; an inside predictor cannot include itself; current technology falls far short
T3. Free will is an illusion	True for libertarian free will, given T1	Compatibilist free will survives; the chooser is the computation that decides
T4a. Black holes to white holes, in cycles	Speculative	Local bounces untested; global recollapse needs weakening dark energy
T4b. About 110 billion years per cycle	No derivation	Would be fixed by measured dark-energy evolution
T4c. Signatures last the entire time	Consistent if and only if dynamics are unitary	Refuted by genuine information loss in black holes or bounces

Table 3. The Fold Hypothesis, claim by claim.

8. Limitations

This is a conceptual paper. Its propositions are elementary and its simulations are toys; their job is to make each claim exact, not to model the universe. The cat map is a finite, classical, perfectly reversible stand-in for unitary dynamics. Real scrambling is quantum and takes place in an astronomically larger state space. Proposition 3(b) assumes each part of the seed is refreshed independently; correlated refreshment changes the rate, not the conclusion. The capacity estimate in Section 2.3 is an order-of-magnitude argument. The cosmology of Section 4 summarizes active research whose observational status may change quickly, and the account of Tootsie Roll manufacture is a popular one that we have not verified. None of this bears on the main result, Theorem 1, which depends only on the definitions of information and entropy.

9. Conclusion

The Fold Hypothesis survives, but not in the form that first suggests itself. A seed fixes the future, and with the exact seed and laws the future could be read off it, as a backtest reads off the past. But the reading must be exact, something must do it, and no reader inside the universe can read its own page. That gap between determining and revealing is where both chaos and choice live. The free will it leaves is not the power to have done otherwise, but the fact that nothing settles a choice before the chooser does. And the persistence of what we were is not a matter of a small portion of each batch going into the next, which would fade as p^n . It is a matter of nothing ever being thrown away. If the universe’s laws are unitary, as most physicists now expect, then everything that has happened is still here, folded beyond recognition into the present, and will be folded into every seed after it. Not a little of the last batch, but all of it: folded, not forgotten.

Data & Materials Availability

The toy universe and the precision horizon are produced by `code/toy.py` (seed 20261003), which writes `results/results.json` and both figures.

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Appendix. Proofs

Proposition 1. Linearizing about the true trajectory, a seed error ε along the most unstable direction grows as $\varepsilon e^{\lambda t}$ until it reaches the scale of the system. Requiring $\varepsilon e^{\lambda t} \leq \delta$ gives $\varepsilon \leq \delta e^{-\lambda t}$, that is, $\log_2(\delta/\varepsilon) \geq \lambda t / \ln 2$ bits. With $\varepsilon = 10^{-d}$ and δ of order 1, $t \approx d \ln 10 / \lambda$. For the cat map the Jacobian is constant, with eigenvalues $(3 \pm \sqrt{5})/2$, so $\lambda = \ln[(3 + \sqrt{5})/2]$.

Proposition 2. The recursion $S_{n+1} = pS_n + a$ has solution $S_n = p^n S_0 + a(1 - p^n)/(1 - p)$ for $p \neq 1$, which tends to $a/(1 - p)$, and $S_n = S_0 + na$ for $p = 1$.

Proposition 3. (a) If $s_n = G^n(s_0)$ for an invertible G , then s_0 is a function of s_n , so $H(s_0 | s_n) = 0$ and $I(s_0; s_n) = H(s_0)$. (b) Undo the reversible scramble, which changes no mutual information. Each part of the seed then survives n cycles with probability p^n , independently, and is otherwise replaced by material independent of s_0 . If the observer can tell which parts are original (an erasure channel), then $I = p^n H(s_0)$ exactly. If not, $I \leq p^n H(s_0)$, and the correlation between a recovered part and the original is p^n when the fresh material has the signature’s own mean, as in Figure 2. (c) The Markov chain $s_0 \rightarrow s_{n-1} \rightarrow s_n$ and the strong data-processing inequality give $I(s_0; s_n) \leq \eta I(s_0; s_{n-1})$ (Ahlsweide and Gács 1976). Induction completes the proof.

Theorem 1. Sufficiency: under a unitary U , the fine-grained state $\rho_{n+1} = U\rho_n U^\dagger$ has constant von Neumann entropy, and $\rho_0 = U^{-n}\rho_n U^n$ is recoverable, so Proposition 3(a) applies. Meanwhile the coarse-grained entropy, computed after projecting onto macroscopic observables, can grow within a cycle and be reduced by a change

of macroscopic description at the bounce, so Proposition 2 with $p < 1$ applies to it. Necessity: if the fine-grained map loses information at any rate $\eta < 1$, Proposition 3(c) sends $I(s_0; s_n)$ to zero. If the same quantity that carries signatures were also the entropy subject to Proposition 2, then $p = 1$ would make entropy diverge, contradicting endless repetition, and $p < 1$ would make signatures fade.